# Systems Analysis and Control

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Lecture 19: Bode Plots, Part 1

# Overview

In this Lecture, you will learn:

**Getting Started with Bode Plots**

- Basic Setup
- Sketching Steps
- Decomposing a transfer function

**Simple Plots**

- Constants
- Adding Real Zeros and Poles

# Review of Frequency Response

## Sinusoidal Input implies Sinusoidal Output (With Amplification and Phase Shift)

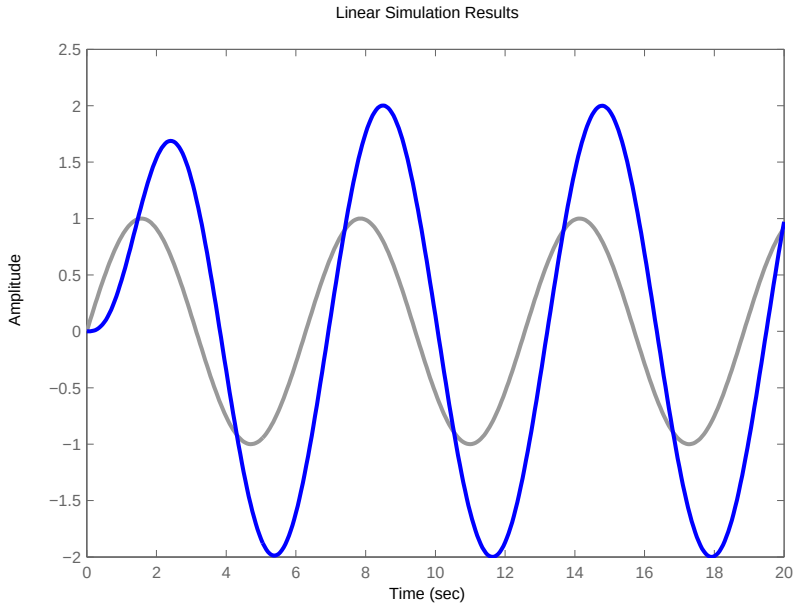
Recall from last lecture: Frequency Response

**Input:**

$$u(t) = M \sin(\omega t + \phi)$$

**Output:** Magnitude and Phase Shift

$$y(t) = |G(\imath\omega)| M \sin(\omega t + \phi + \angle G(\imath\omega))$$

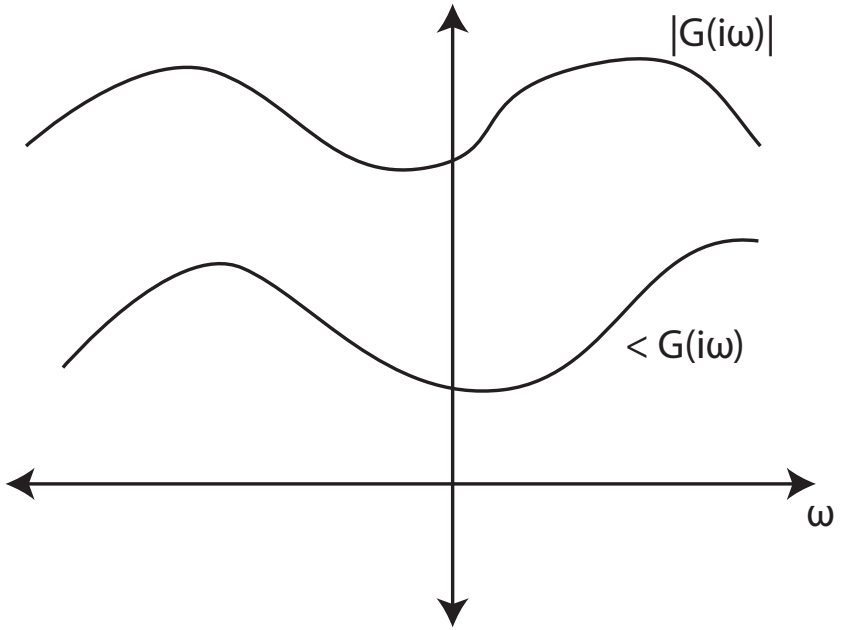



Frequency Response to $\sin \omega t$ is given by $G(\imath\omega)$

# Bode Plots: Visualising the Frequency Response

Visualizing $|G(\imath\omega)|$ and $\angle G(\imath\omega)$:

- 1 independent Variable: $\omega$
- 2 Dependent Variables: $\angle G(\imath\omega)$ and $|G(\imath\omega)|$




- Advantage: All Information corresponds to physical data.
  - Can be found directly using a frequency sweep.

**Problem:** Getting all important information into one plot!

- $|G(\imath\omega)|$ gets very small and very big
- low frequency information gets crowded out.

# Bode Plots: Use Log Scaling for $\omega$ and $|G(\imath\omega)|$

We measure $|G(\imath\omega)|$ in Decibels

**Definition 1.**

**The Bode Plot** is a pair of $\log$-$\log$ and semi-$\log$ plots:

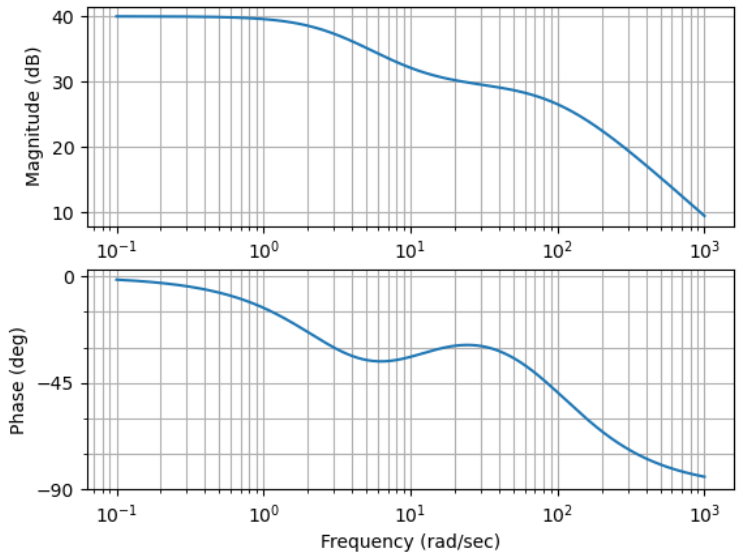
1. Magnitude Plot: $20\log_{10}|G(\imath\omega)|$ vs. $\log_{10}\omega$
2. Phase Plot: $\angle G(\imath\omega)$ vs. $\log_{10}\omega$

$20\log_{10}|G(\imath\omega)|$ is units of **Decibels (dB)**

- Used in Power and Circuits.
- $10\log_{10}|\cdot|$ in other fields.

Note that by $\log$, we mean $\log$ base 10 ($\log_{10}$)

- In Matlab, `log` means natural logarithm.

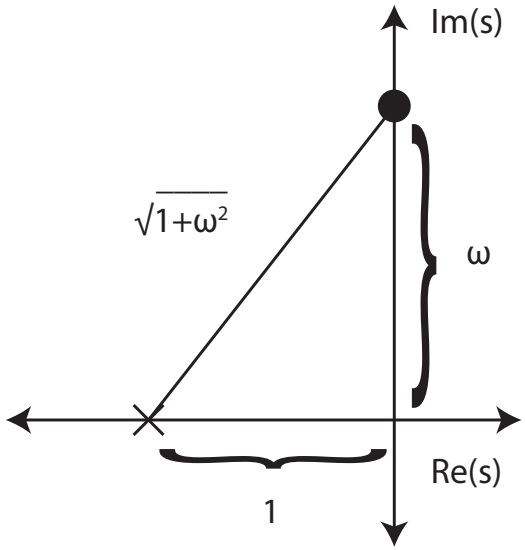


# Bode Plot of a Single Pole

## Analytic Construction of Gain, $|G(\imath\omega)|$

To illustrate the process, lets Bode plot a simple pole at $s = -1$

$$G(s) = \frac{1}{s+1}$$




We need

- Magnitude of $G(\imath\omega)$
- Phase of $G(\imath\omega)$

Recall that

$$|G(s)| = \frac{|s - z_1| \cdots |s - z_m|}{|s - p_1| \cdots |s - p_n|}$$

So that

$$|G(\imath\omega)| = \frac{1}{|\imath\omega + 1|} = \frac{1}{\sqrt{1 + \omega^2}}$$

# Bode Plots: How to Bode Plot $|G(\imath\omega)| = \frac{1}{\sqrt{1+\omega^2}}$?

For Log-Log format, we divide the frequency into 3 regions

How to Plot $|G(\imath\omega)| = \frac{1}{\sqrt{1+\omega^2}}$?

We actually want to plot it in dB, so ...

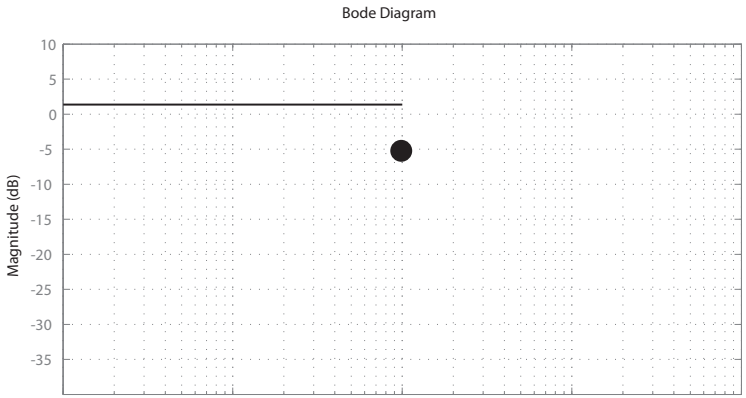
$$20\log|G(\imath\omega)| = 20\log\frac{1}{\sqrt{1+\omega^2}} = 20\log(1+\omega^2)^{-\frac{1}{2}} = -10\log(1+\omega^2)$$

Three Cases:

**Case 1:** $\omega << 1$

- Approximate $1+\omega^2 \cong 1$

$$\begin{aligned} 20\log|G(\imath\omega)| &= -10\log(1+\omega^2) \\ &\cong -10\log 1 = 0 \end{aligned}$$




**Case 2:** $\omega = 1$

$$\begin{aligned} 20\log|G(\imath\omega)| &= -10\log(1+\omega^2) \\ &= -10\log(1+\omega^2) \\ &= -3.01 \end{aligned}$$

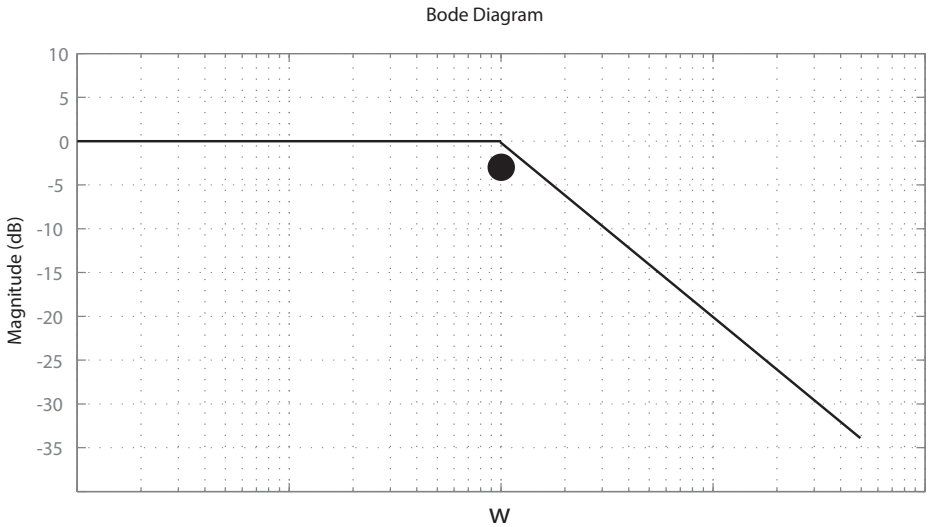
# In Log-Log plots, $|G(\imath\omega)| = \frac{1}{\sqrt{1+\omega^2}}$ looks Piecewise-Linear

## Find Slope and Inflection Point

**Case 3:** $\omega >> 1$

- Approximate: $1 + \omega^2 \cong \omega^2$

$$\begin{aligned} 20\log|G(\imath\omega)| &= -10\log(1+\omega^2) \\ &\cong -10\log\omega^2 \\ &= -20\log\omega \end{aligned}$$




But we use a $\log - \log$ plot.

- $x$-axis is $x = \log\omega$
- $y$-axis is $y = 20\log|G(\imath\omega)| = -20\log\omega = -20x$

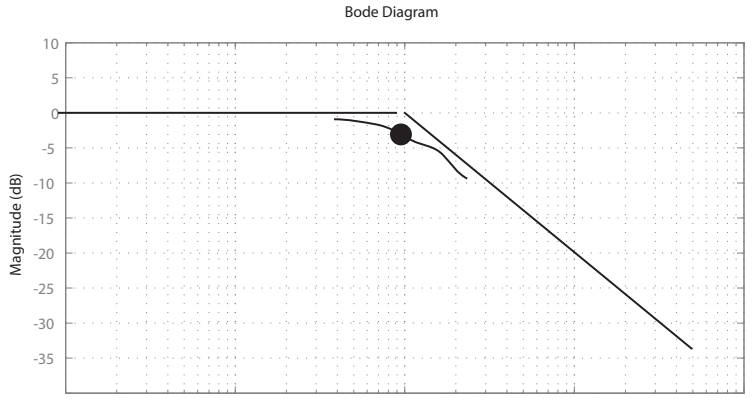
**Conclusion:** On the $\log$-$\log$ plot, when $\omega >> 1$,

- Plot is Linear
- Slope is -20 dB/Decade!

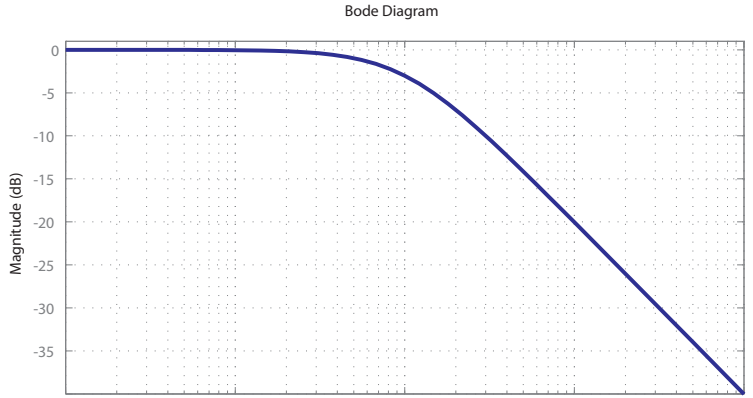
# Bode Plot: Connecting at the Inflection Point

## Comparison with Matlab

Of course, we need to connect the dots.




Compare to the Real Thing (From Matlab: `bode(sys)`):



# Bode Plot of a Single Pole

## Analytic Construction of Phase, $\angle G(\imath\omega) = -\tan^{-1}(\omega)$

Now lets do the phase. Recall:

$$\angle G(s) = \sum_{i=1}^{m} \angle(s - z_i) - \sum_{i=1}^{n} \angle(s - p_i)$$

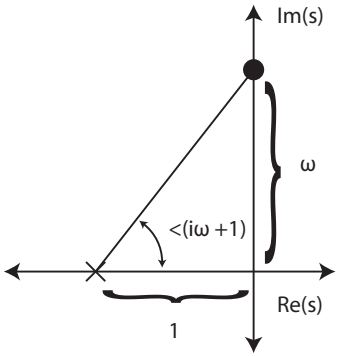
In this case, forming a triangle, we have

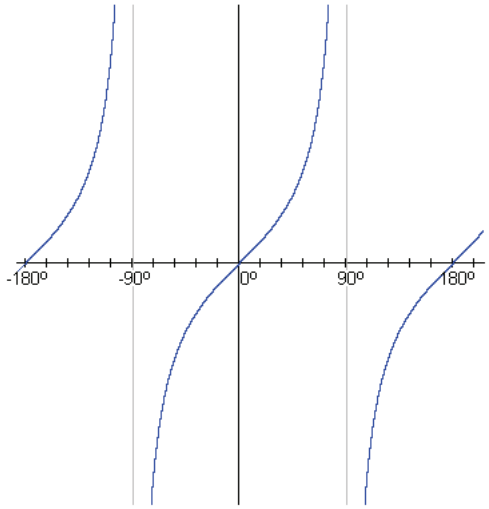
$$\begin{aligned} \angle G(\imath\omega) &= -\angle(\imath\omega + 1) \\ &= -\tan^{-1}(\omega) \end{aligned}$$

Again, 3 cases:

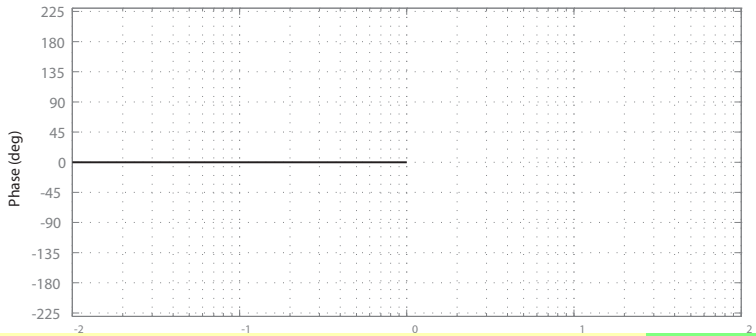
**Case 1:** $\omega << 1$

- $\tan(\angle G(\imath\omega)) \cong 0$
- $\tan(\angle G(\imath\omega)) \cong \angle G(\imath\omega) \cong 0$




Graph of the Tangent Function



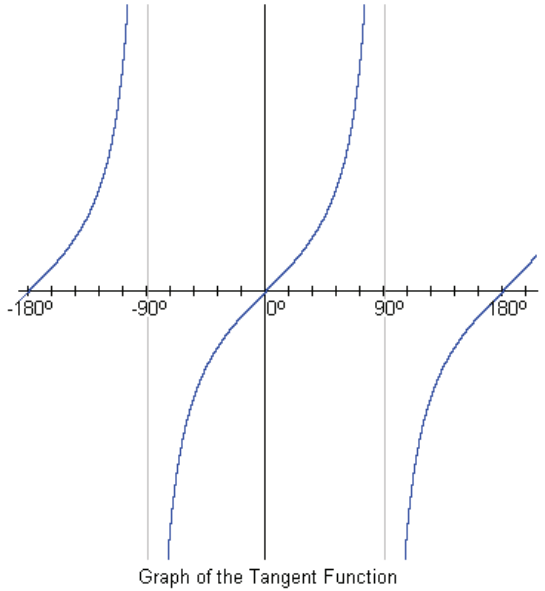
# Bode Plot of $\angle G(\imath\omega) = -\tan^{-1}(\omega)$

Inflection points occur at $\omega = .1$ and $\omega = 10$

**Case 2:** $\omega = 1$

- $\tan(\angle G(\imath\omega)) = 1$
- $\angle G(\imath\omega) \cong 45°$

**Problem:** $\tan^{-1}(\omega)$ is smooth at $\omega = 1$



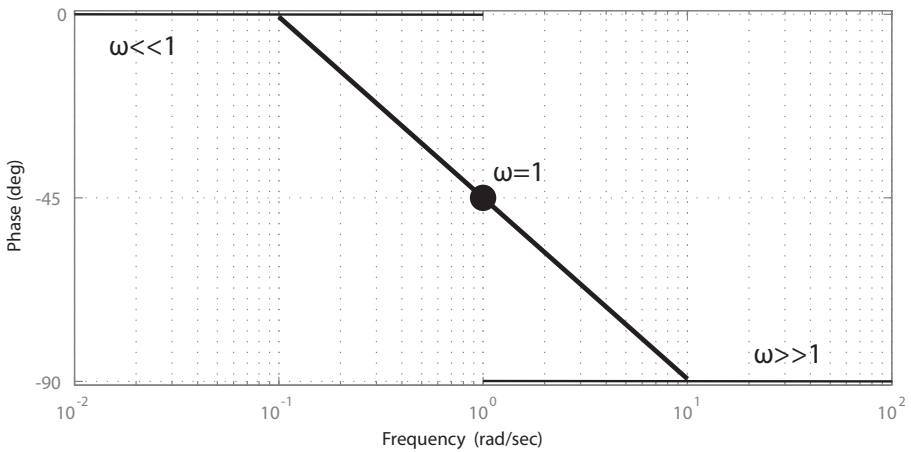
Graph of the Tangent Function

**Between** $\omega = .1$ **and** $\omega = 10$**:**

- Approximate Slope:
  - $-45°$/Decade
- Two inflection points!

**Case 3:** $\omega >> 1$

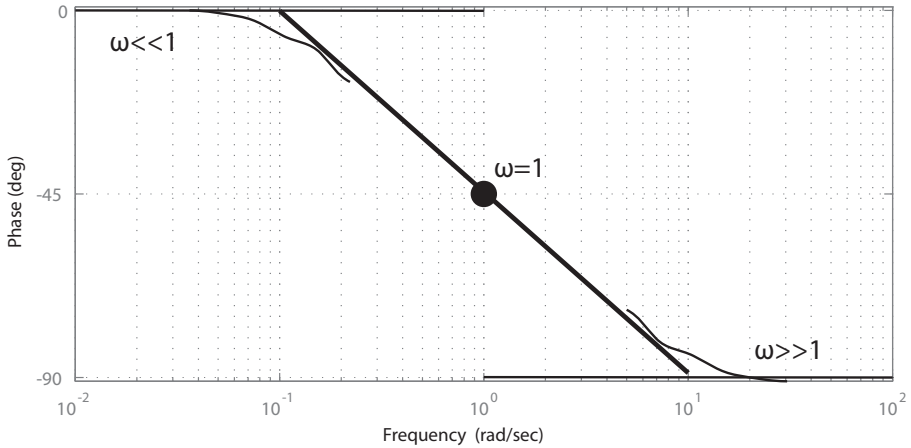
- $\tan(\angle G(\imath\omega)) \cong \frac{1}{0}$
- $\angle G(\imath\omega) \cong -90°$
- Fixed at $-90°$ for large $\omega$!



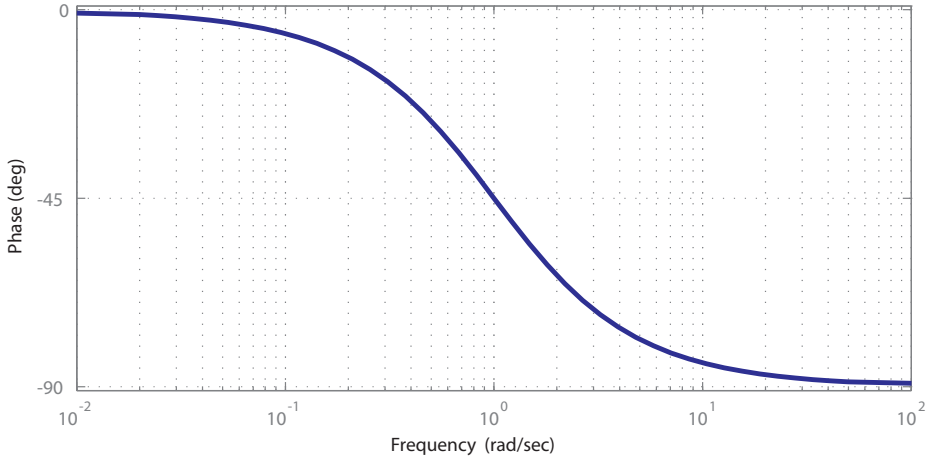
# Bode Plot: Connecting at the Inflection Points

## Comparison with Matlab

We need to connect the dots somehow.




Compare to the real thing:



# General Method For Sketching Bode Plots

## Rule 1: Magnitude Plots Add

So far, drawing Bode Plots seems pretty intimidating.

- Solving $\tan^{-1}$
- dB and $\log$-plots
- Lots of trig

The process can be Greatly Simplified:

- Use a few simple rules.

**Example:** Suppose we have

$$G(s) = G_1(s)G_2(s)$$

Then

$$|G(\imath\omega)| = |G_1(\imath\omega)||G_2(\imath\omega)|$$

and

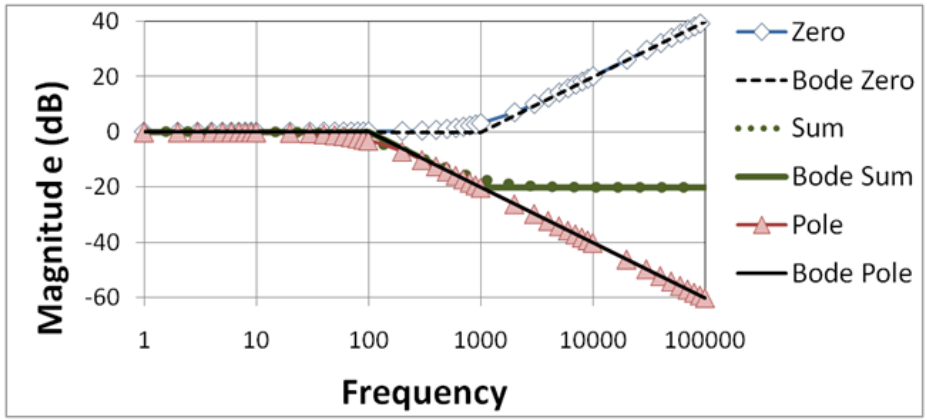
$$\log|G(\imath\omega)| = \log|G_1(\imath\omega)| + \log|G_2(\imath\omega)|$$

# General Method For Sketching Bode Plots

## Rule 1: Magnitude Plots Add

**Rule # 1:** Magnitude Plots Add in log-space.
For $G(s) = G_1(s)G_2(s)$,

$$20 \log |G(\imath\omega)| = 20 \log |G_1(\imath\omega)| + 20 \log |G_2(\imath\omega)|$$




**Note:** constant gains also add to the gain plot:

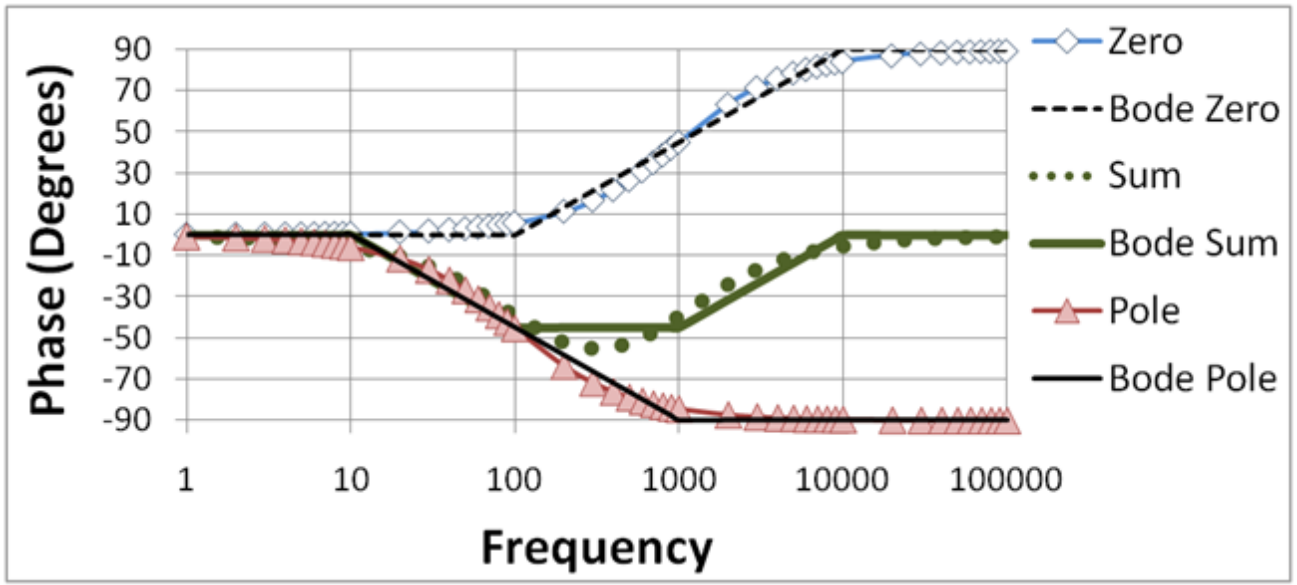
$$20 \log |kG(\imath\omega)| = 20 \log k + 20 \log |G(\imath\omega)|$$

# General Method For Sketching Bode Plots

## Rule 1: Magnitude Plots Add

**Rule # 2:** Phase Plots Add.
For $G(s) = G_1(s)G_2(s)$,

$$\angle G(\imath\omega) = \angle G_1(\imath\omega) + \angle G_2(\imath\omega)$$



# General Method For Sketching Bode Plots

## Step 1: Decompose $G(s)$ into isolated poles/zeros

Our Approach is to Decompose $G(s)$ into simpler pieces.

- Plot the phase and magnitude of each component.
- Add up the plots.

**Step 1:** Decompose $G$ into all its poles and zeros

$$G(s) = \frac{(s - z_1) \cdots (s - z_m)}{(s - p_1) \cdots (s - p_n)}$$

Then for magnitude

$$\begin{aligned} 20 \log |G(\imath\omega)| &= \sum_i 20 \log |\imath\omega - z_i| + \sum_i 20 \log \frac{1}{|\imath\omega - p_i|} \\ &= \sum_i 20 \log |\imath\omega - z_i| - \sum_i 20 \log |\imath\omega - p_i| \end{aligned}$$

And for phase:

$$\angle G(\imath\omega) = \sum_i \angle(\imath\omega - z_i) - \sum_i \angle(\imath\omega - p_i)$$

But how to plot $\angle(\imath\omega - z_i)$ and $20 \log |\imath\omega - z_i|$?

General Method For Sketching Bode Plots
Step 1: Decompose $G(s)$ into isolated poles/zeros

Our Approach is to Decompose $G(s)$ into simpler pieces.
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**Step 1:** Decompose $G$ into all its poles and zeros

$$G(s) = \frac{(s-z_1)\cdots(s-z_m)}{(s-p_1)\cdots(s-p_n)}$$

Then for magnitude

$$20\log|G(\imath\omega)| = \sum_i 20\log|\imath\omega - z_i| + \sum_i 20\log\frac{1}{|\imath\omega - p_i|}$$
$$= \sum_i 20\log|\imath\omega - z_i| - \sum_i 20\log|\imath\omega - p_i|$$

And for phase:

$$\angle G(\imath\omega) = \sum_i \angle(\imath\omega - z_i) - \sum_i \angle(\imath\omega - p_i)$$

But how to plot $\angle(\imath\omega - z_i)$ and $20\log|\imath\omega - z_i|$?

Decompose $G$ into bite-size chunks: isolated real poles/zeros and complex poles/zeros

$$G(s) = \frac{1}{s+3}(s+1)\frac{1}{s^2+3s+1} = G_1(s)G_2(s)G_3(s)$$

# Step 2: Normalize the Terms

## Isolate the Low-Frequency Gain

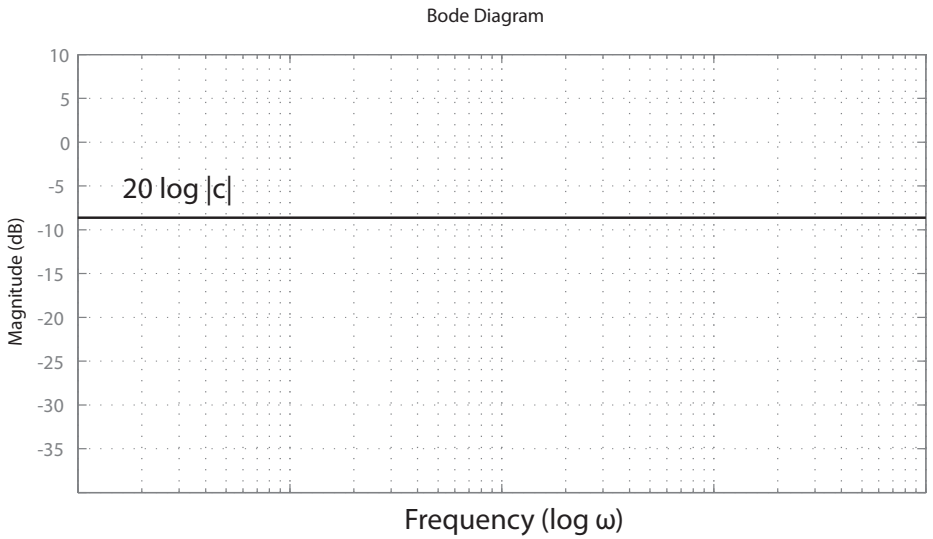
**Step 2:** Normalize all isolated poles/zeros so they have no low-frequency gain

$$G(s) = k\frac{(s-z_1)\cdots(s-z_m)}{(s-p_1)\cdots(s-p_n)} = \underbrace{k\frac{z_1\cdots z_m}{p_1\cdots p_n}}_{c}\frac{(\frac{s}{z_1}-1)\cdots(\frac{s}{z_m}-1)}{(\frac{s}{p_1}-1)\cdots(\frac{s}{p_n}-1)}$$

**Recall:** We also did this when computing gains on the root-locus

**Plot and add** $20\log|c|$**:** just like any other term

- $|G_0(\imath\omega)| = |c|$




Conclusion: Bode plot of a contant, $c$, is Constant for all $\omega$

# Why Normalize the Terms?

Steady-State response is not just $y_{ss} = c$

$$G(s) = c\frac{(\tau_{z1}s + 1)\cdots(\tau_{zm}s + 1)}{(\tau_{p1}s + 1)\cdots(\tau_{pn}s + 1)}$$

The advantage of this form is that steady-state response to a step is

$$y_{ss} = \lim_{s\to 0} G(s) = G(0) = c$$

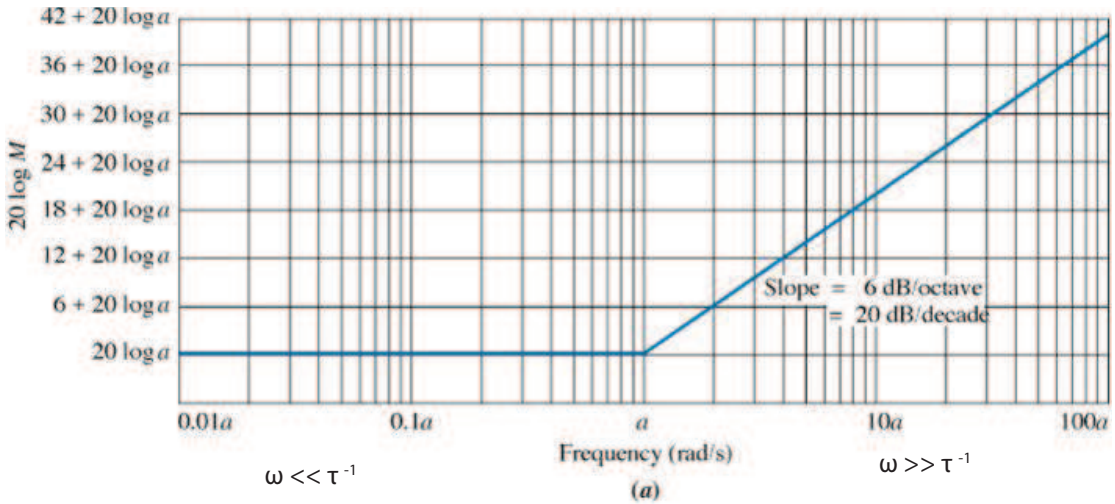



Figure: Gain Plot of $G(s) = s + a$

Low Frequency Response is given by the constant term, $c$.

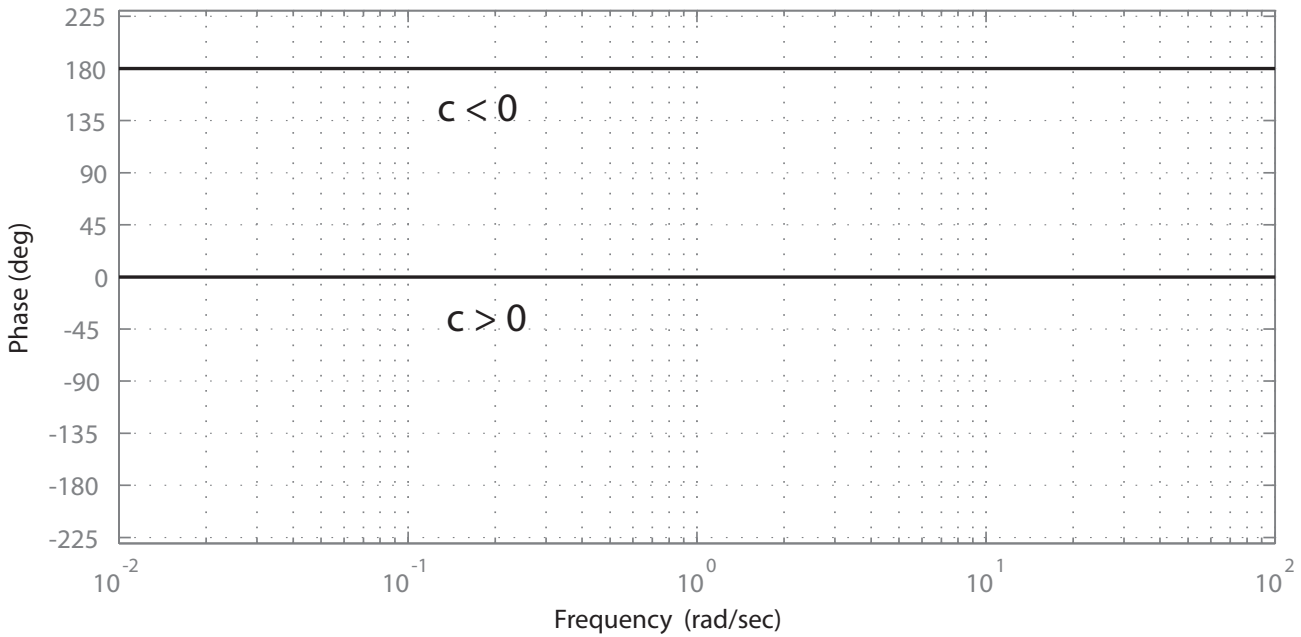
# Question: Does the Constant have a phase plot?

Answer: Yes, but its just $\angle c = 0$ or $\angle c = 180°$

**Step 3:** Plot the magnitude and phase of the gain term: $G_0(s) = c$.

**Phase of a Constant:** $G_0(s) = c$

$$\angle G_0(\imath\omega) = \angle c = \begin{cases} 0° & c > 0 \\ 180° & c < 0 \end{cases}$$




Conclusion: phase is $0°$ if $c > 0$, otherwise $180°$.

# Step 3: Plot the "Pure" Zeros – $G_i(s) = s$

Gain plot: Pure Zeros have no inflection points, but a y-intercept at $\omega = 1$

Lets start with a zero at the origin: $G_1(s) = s$.

**Magnitude:** $G_1(s) = s$

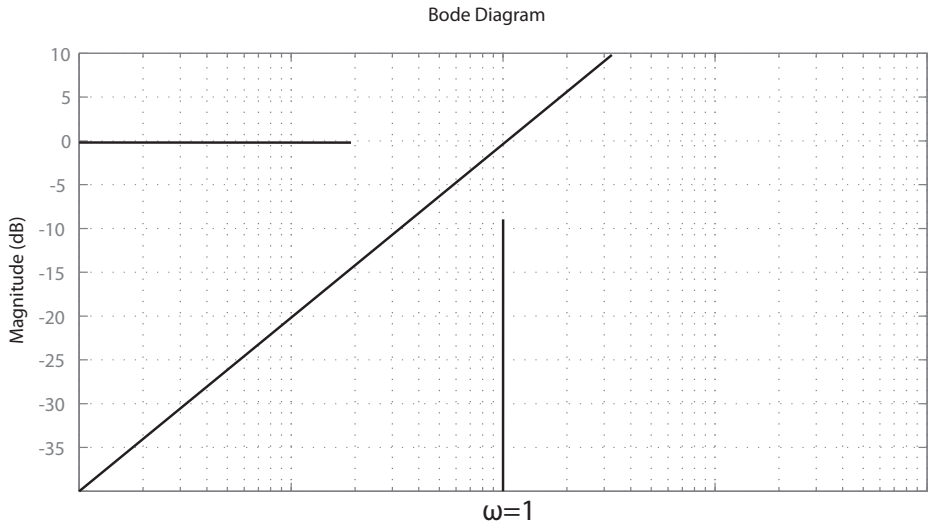
- $|G_1(\imath\omega)| = |\imath\omega| = |\omega|$
- $20\log|G_1(\imath\omega)| = 20\log|\omega|$

Our x-axis is $\log\omega$.

- Plot is Linear for all $\omega$
- Slope is $+20$ dB/Decade!
- Need a point: $\omega = 1$

$$20\log|G_1(\imath\omega)||_{\omega=1} = 20\log 1 = 0$$

- Passes through 0dB at $\omega = 1$




High Gain at High Frequency

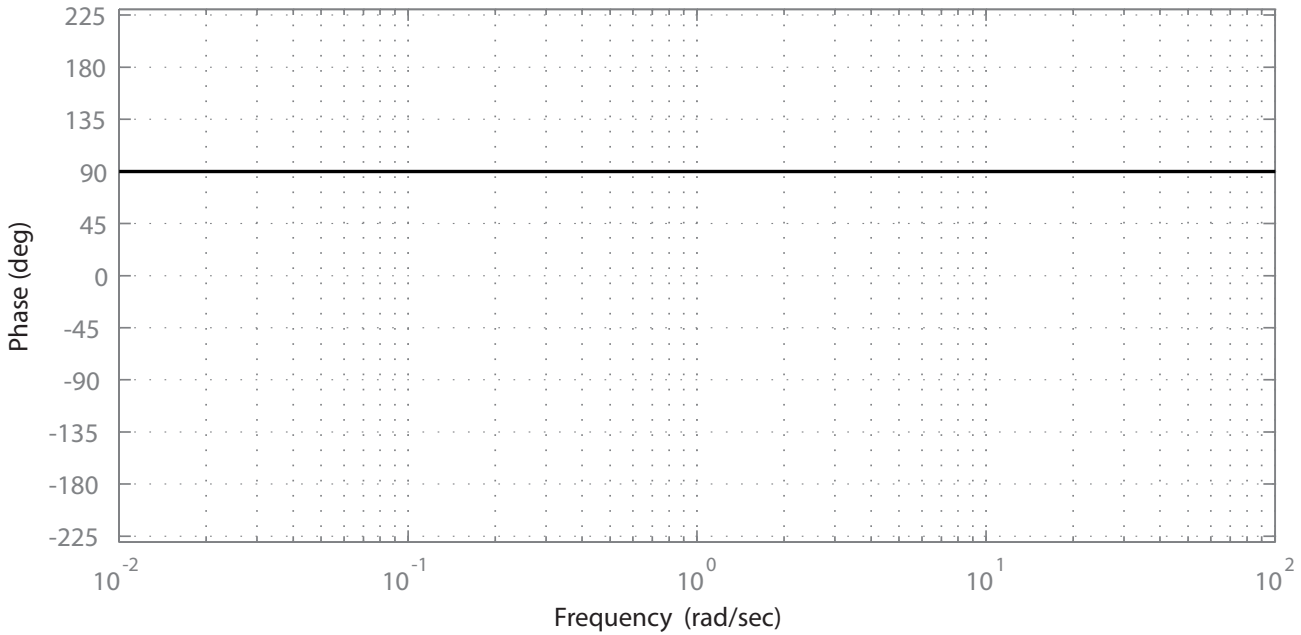
- A pure zero means $u'(t)$
- The faster the input, The larger the output

# Step 3: Plot the "Pure" Zeros – $G_i(s) = s$

## Phase plot: Constant phase – $\angle G_i(\imath\omega) = \angle \imath\omega = 90^\circ$

**Phase:** $G_i(s) = s$

- $\angle G_i(\imath\omega) = \angle \imath\omega = 90^\circ$
- Always $90^\circ$!




Always $90^\circ$ out of phase. Why?

# Combine Multiple "Pure" Zeros – $G_i(s) = s^k$

Gain plot: y-intercept still at $\omega = 1$, slope increases to $20 \cdot k$ dB/dec

What happens if there are multiple pure zeros

- Just what you would expect.

**Magnitude:** $G_1(s) = s^k$

- $|G_1(\imath\omega)| = |\imath\omega|^k = |\omega|^k$

$$20 \log |G_1(\imath\omega)| = 20 \log |\omega|^k$$
$$= 20k \log |\omega|$$

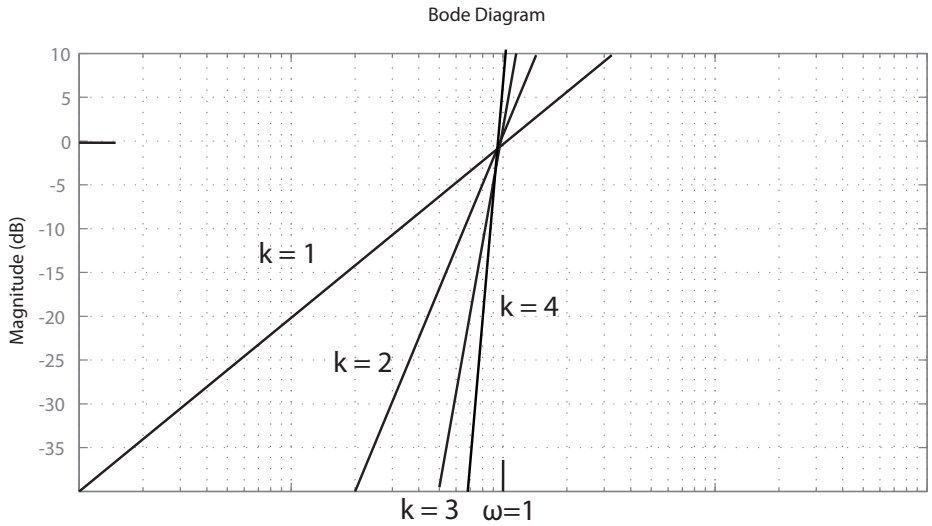
- Slope is +20k dB/Decade!

**Need a Point**

- At $\omega = 1$:

$$20 \log |G_1(\imath\omega)||_{\omega=1} = 20k \log 1 = 0$$

- Still Passes through $0dB$ at $\omega = 1$




$k$ pure zeros added together.

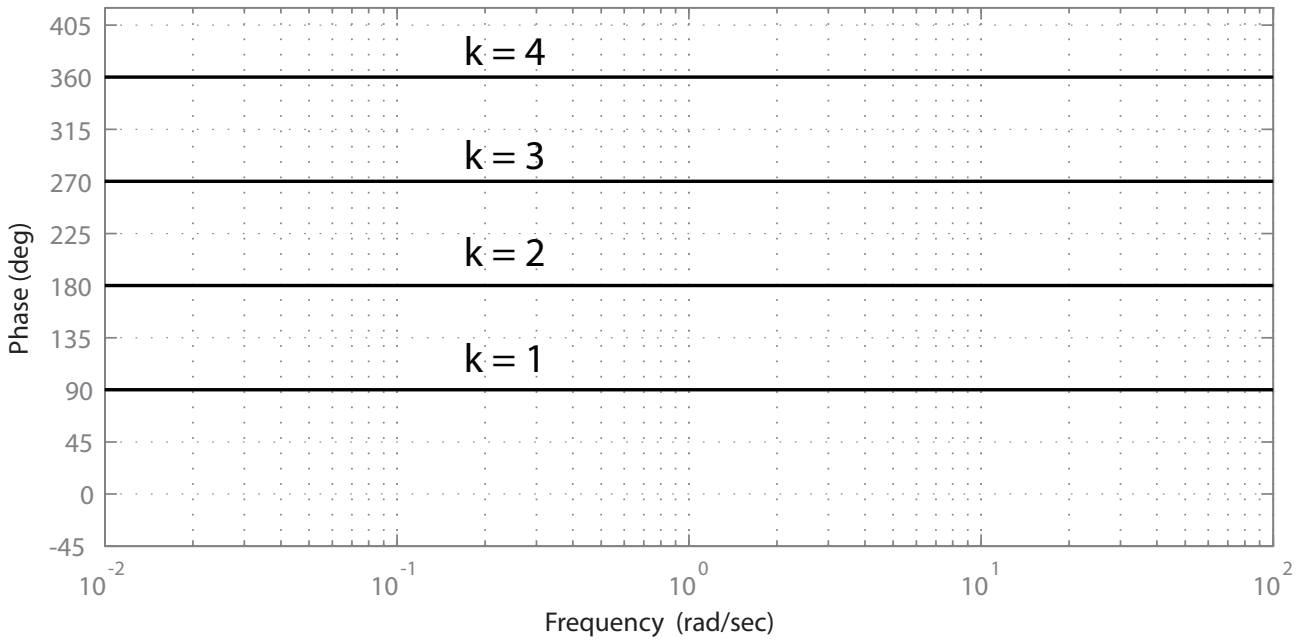
# Combine Multiple "Pure" Zeros – $G_i(s) = s^k$

Phase plot: Still constant phase – but $\angle(\imath\omega)^k = k \cdot 90°$

And phase for multiple pure zeros?

**Phase:** $G_i(s) = s^k$

- $\angle G_i(\imath\omega) = \angle(\imath\omega)^k = k\angle\imath\omega = 90°k$
- Always $90°k$




$k$ pure zeros added together.

# Step 4: Plot the Other Zeros – $G_i(s) = (s - z_i)$

First: An annoying change in notation

**Recall:** We already plotted $G_i(s) = \frac{1}{s+1}$.

- Real zeros are similar

Change of Format: to simplify expression of steady-state response, we write

$$G_i(s) = \frac{1}{z_i}s + 1 = \tau s + 1$$

- Zero is at $s = -\frac{1}{\tau}$
- Also put poles in this form

**Rewrite $G(s)$:** using $\tau_{z_i}$ and $\tau_{p_i}$

$$G(s) = c\frac{(\frac{1}{z_1}s + 1)\cdots(\frac{1}{z_m}s + 1)}{(\frac{1}{p_1}s + 1)\cdots(\frac{1}{p_n}s + 1)}$$
$$= c\frac{(\tau_{z1}s + 1)\cdots(\tau_{zm}s + 1)}{(\tau_{p1}s + 1)\cdots(\tau_{pn}s + 1)}$$

Where

- $\tau_{zi} = \frac{1}{z_i}$
- $\tau_{pi} = \frac{1}{p_i}$
- $c = k\frac{z_1\cdots z_m}{p_1\cdots p_n}$

Assume $z_i$ and $p_i$ are *Real*.

# Step 4: Plot the Other Zeros – $G_i(s) = (s - z_i)$

**Question:** Why are we not doing pure poles?
**Answer:** As we will see, plotting poles is the same as zeros, but Bode plot is flipped. So we are doing all the zeros first.

Step 4: Plot the Other Zeros – $G_i(s) = (s - z_i)$
First: An annoying change in notation

**Recall:** We already plotted $G_i(s) = \frac{1}{s+1}$.
- Real zeros are similar

Change of Format: to simplify expression of steady-state response, we write

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- Zero is at $s = -\frac{1}{\tau}$
- Also put poles in this form

**Rewrite** $G(s)$: using $\tau_{z_i}$ and $\tau_{p_i}$

$$G(s) = c\frac{(\frac{1}{z_1}s+1)\cdots(\frac{1}{z_m}s+1)}{(\frac{1}{p_1}s+1)\cdots(\frac{1}{p_n}s+1)} = c\frac{(\tau_{z1}s+1)\cdots(\tau_{zm}s+1)}{(\tau_{p1}s+1)\cdots(\tau_{pm}s+1)}$$

Where
- $\tau_{zi} = \frac{1}{z_i}$
- $\tau_{pi} = \frac{1}{p_i}$
- $c = k\frac{z_1\cdots z_m}{p_1\cdots p_n}$

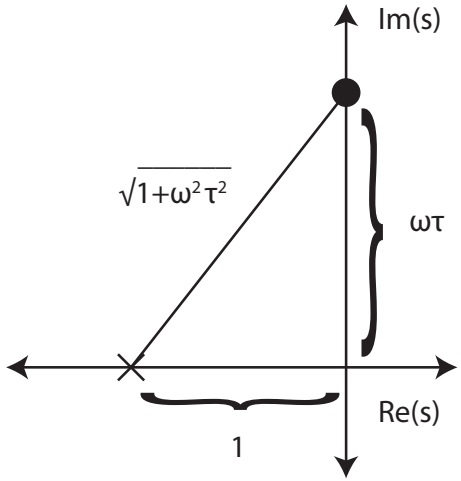
Assume $z_i$ and $p_i$ are *Real*.

# Bode Plots of Real Zero: $|G(\imath\omega)| = \sqrt{1+\tau^2\omega^2}$

For Log-Log format, inflection point at $\omega\tau = 1$

$$G_i(s) = (\tau s + 1)$$

$$|G_i(\imath\omega)| = |\imath\omega\tau + 1| = \sqrt{1+\tau^2\omega^2}$$




**Magnitude:**

$$20\log|G_1(\imath\omega)| = 20\log(1+\omega^2\tau^2)^{\frac{1}{2}} = 10\log(1+\omega^2\tau^2)$$

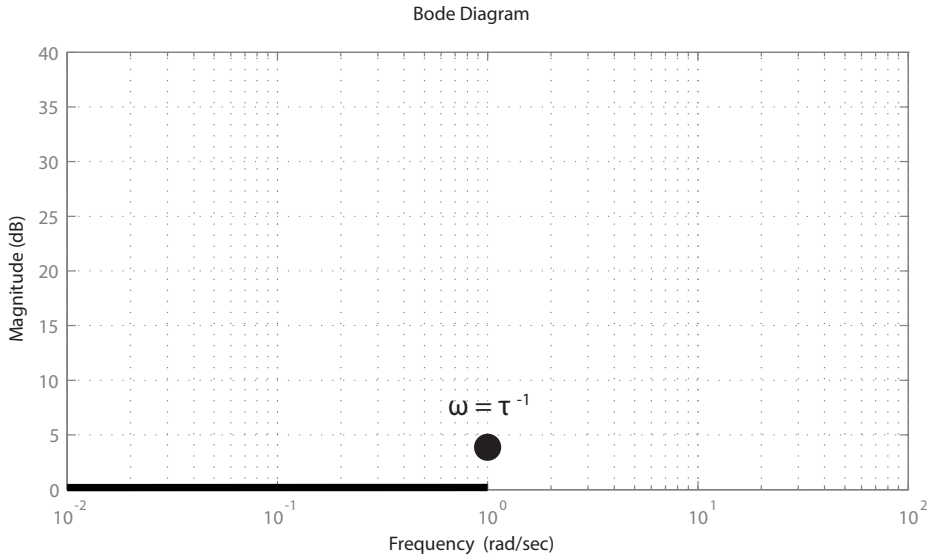
**Case 1:** $\omega\tau << 1$

- Approximate $1+\omega^2\tau^2 \cong 1$

$$20\log|G(\imath\omega)| = 20\log(1+\omega^2\tau^2) \cong 20\log 1 = 0$$

**Case 2:** $\omega\tau = 1$

$$20\log|G(\imath\omega)| = 10\log(1+\omega^2\tau^2) = 10\log 2 = 3.01$$



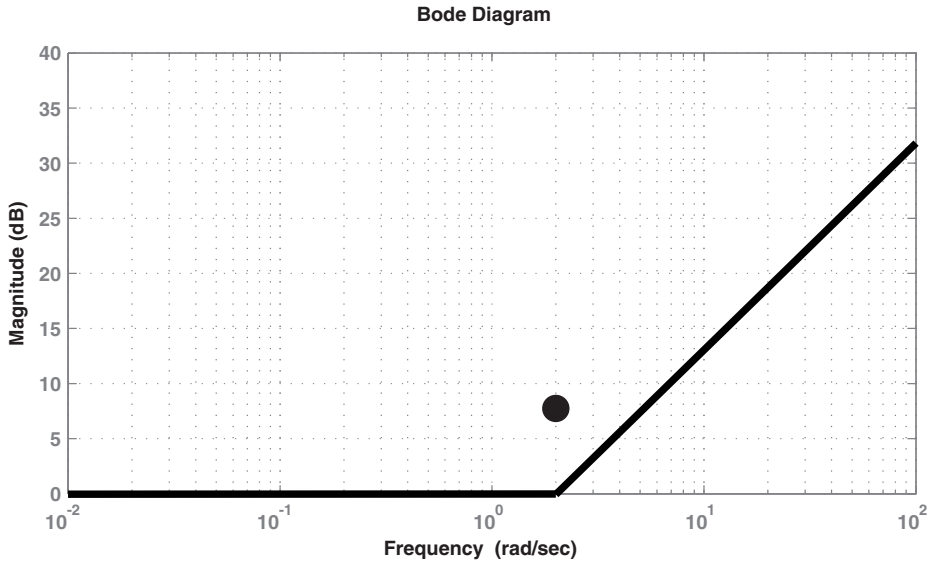
# Bode Plots of Real Zero: $|G(\imath\omega)| = \sqrt{1+\tau^2\omega^2}$

For Log-Log format, high-frequency slope of 20 dB/dec

**Case 3:** $\omega\tau >> 1$

- Approximate $1 + \omega^2\tau^2 \cong \omega^2\tau^2$

$$\begin{aligned} 20\log|G(\imath\omega)| &= 20\log\sqrt{1+\omega^2\tau^2} \\ &\cong 10\log\omega^2\tau^2 \\ &= 20\log\omega\tau \\ &= 20\log\omega + 20\log\tau \end{aligned}$$




**Conclusion:** When $\omega\tau >> 1$,

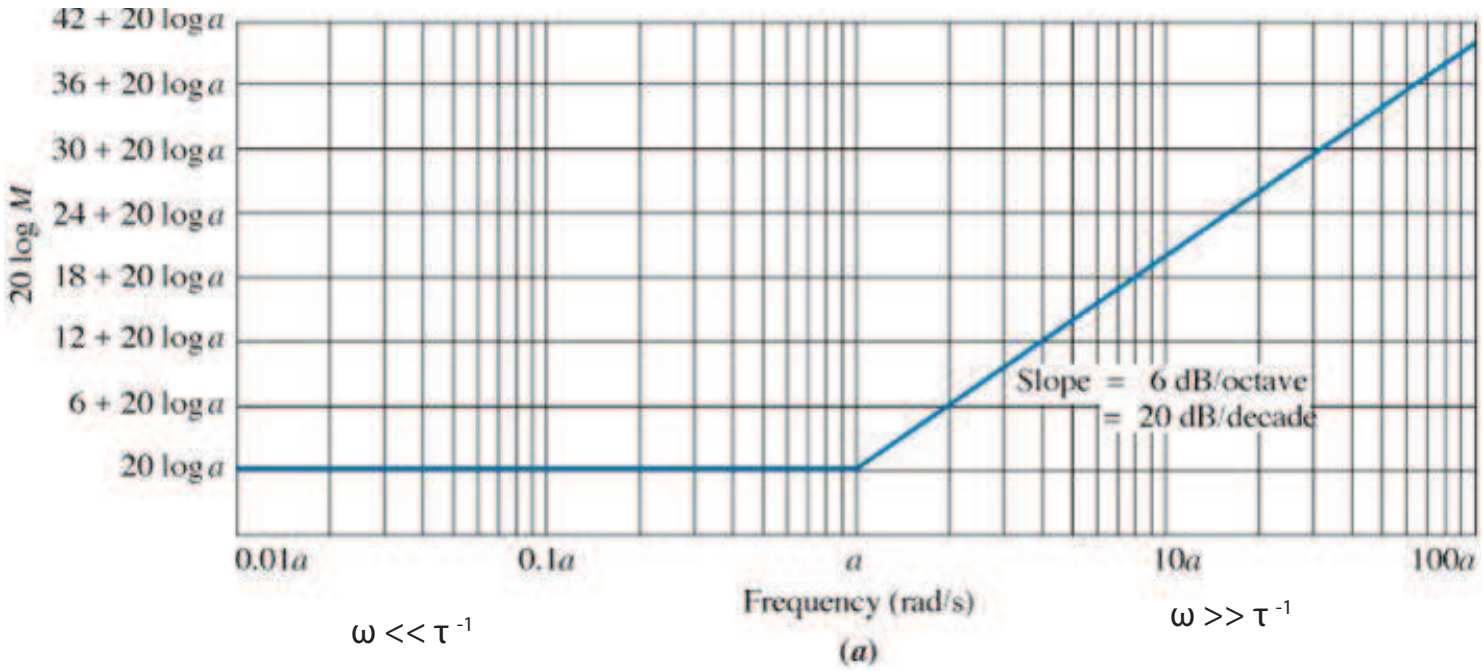
- Plot is Linear
- Slope is +20 dB/Decade!
- inflection at $\omega = \frac{1}{\tau} = z$

# Make sure you normalize!

## Otherwise the magnitude sketch won't work!

Compare this to the magnitude plot of

$$G_i(s) = s + a$$




This is why we use the format $G_1(s) = \tau s + 1$

- We want $0$dB (no gain) at low frequency.

# Gain Plot of Real Zero: $|G(\imath\omega)| = \sqrt{1+\tau^2\omega^2}$

## Summary

$$G_1(s) = (\tau s + 1)$$

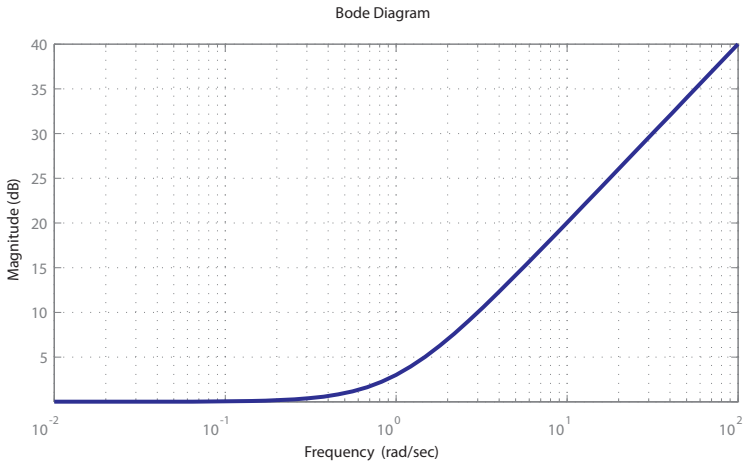
$$|\imath\omega\tau + 1| \cong \begin{cases} 1 & \omega << \frac{1}{\tau} \\ |\imath\omega\tau| & \omega >> \frac{1}{\tau} \end{cases}$$

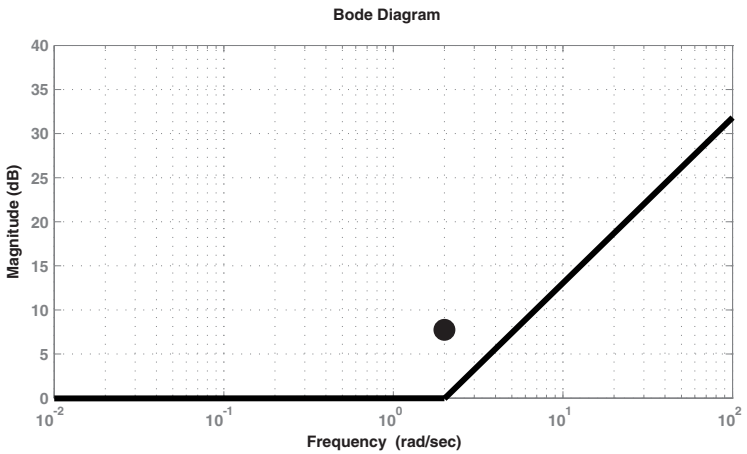
Behaves as

- A constant at low $\omega$.
- A zero at high $\omega$.

**Break Point** at $\omega = \frac{1}{\tau}$.
**Magnitude:**

$$20\log|\imath\omega\tau + 1| = \begin{cases} 0 \text{ dB} & \omega << \frac{1}{\tau} \\ 3.01 \text{ dB} & \omega = \frac{1}{\tau} \\ 20\log\omega & \omega >> \frac{1}{\tau} \end{cases}$$



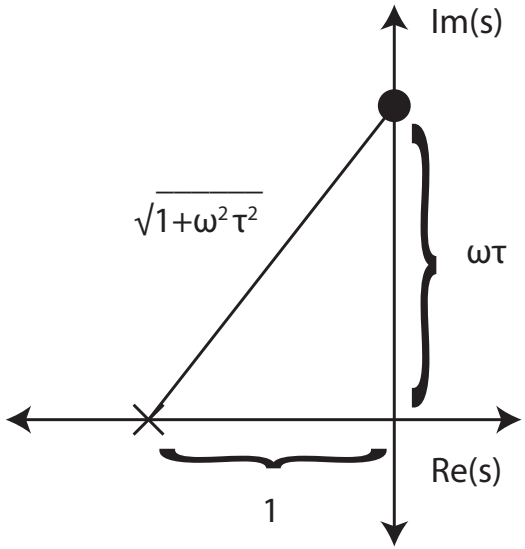
# Phase Plot of Real Zero: $\angle G(\imath\omega) = \tan^{-1}(\omega\tau)$

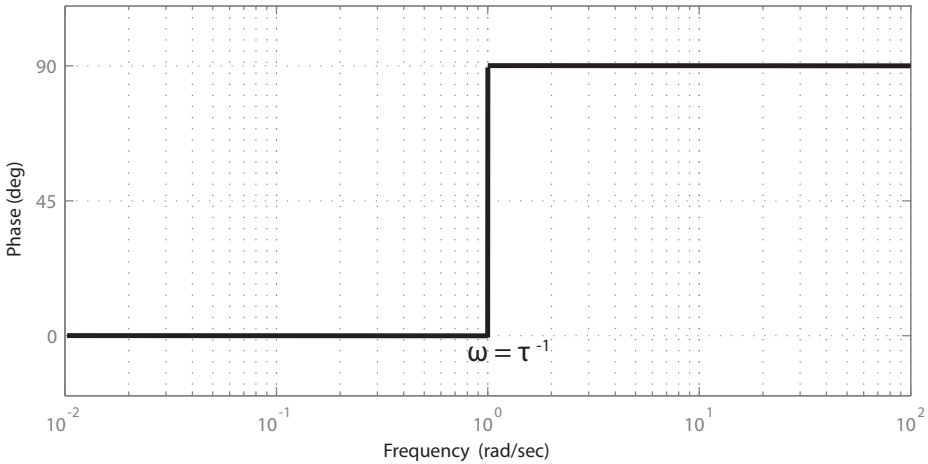
Where to put inflection points?

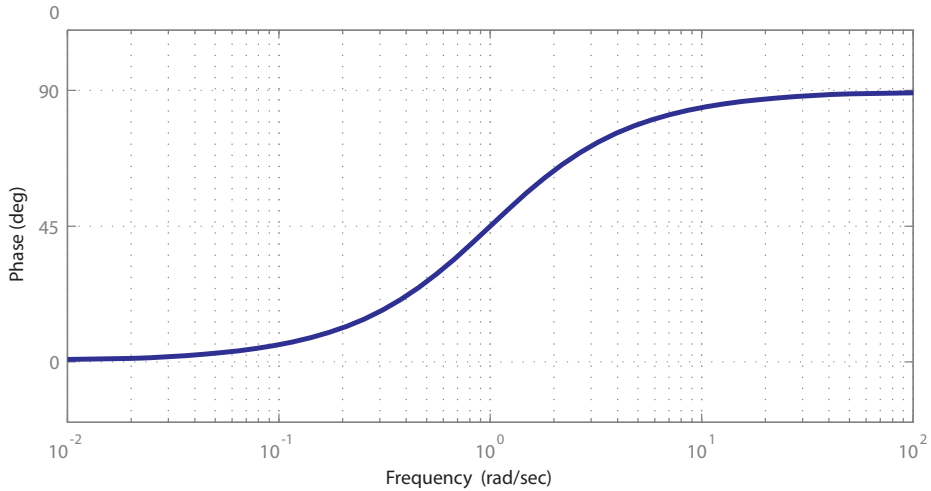
**Geometry:**

$$\angle(\imath\omega\tau + 1) = \tan^{-1}\left(\frac{\omega\tau}{1}\right)$$

$$\angle(\imath\omega\tau + 1) = \begin{cases} 0^\circ & \omega << \frac{1}{\tau} \\ 45^\circ & \omega = \frac{1}{\tau} \\ 90^\circ & \omega >> \frac{1}{\tau} \end{cases}$$

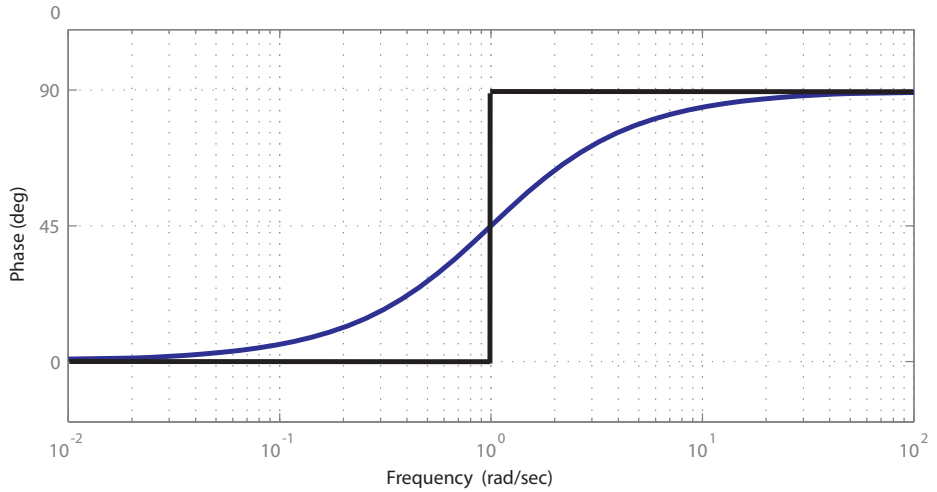


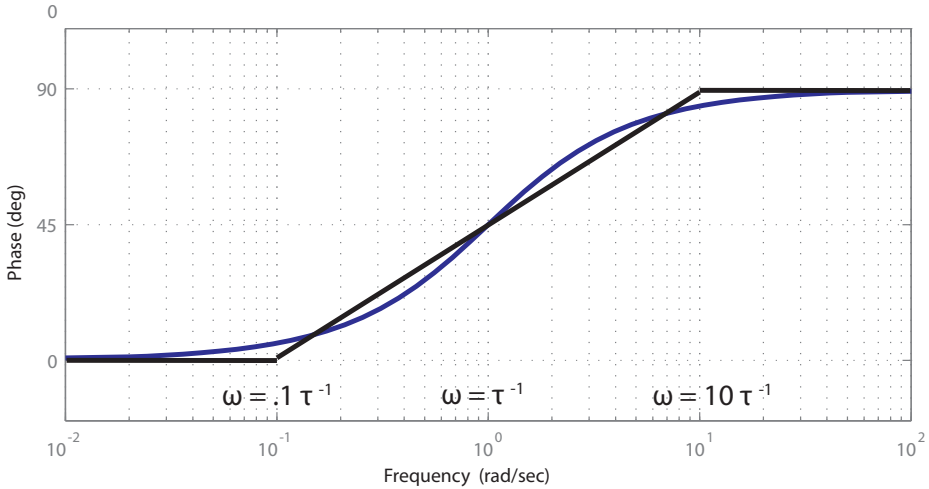
# Phase Plot of Real Zero ($z = \frac{1}{\tau}$): $\angle G(\imath\omega) = \tan^{-1}(\omega\tau)$

Inflection points at $\omega = \frac{1}{10}\frac{1}{\tau}$ and $\omega = 10\frac{1}{\tau}$




We can improve our sketch using a line

$$\angle(\imath\omega\tau + 1)$$
$$= \begin{cases} 0^\circ & \omega < \frac{.1}{\tau} \\ 45^\circ\,(1 + \log\omega\tau) & \frac{.1}{\tau} < \omega < \frac{10}{\tau} \\ 90^\circ & \omega > \frac{10}{\tau} \end{cases}$$




**Remember:** $\frac{1}{\tau} = z$ – Yes, the $\tau$ is unnecessarily complicated.

# Summary

What have we learned today?

**Drawing Bode Plots**

- Drawing Rules

**Simple Plots**

- Constants
- Real Zeros

**Next Lecture: More Bode Plotting**